

MATH 2211, SPRING 2026: HOMEWORK 1

Notation 1. We will use *summation notation*: The symbol

$$\sum_{k=1}^n a_k$$

has to be read like a for loop: The entries $a_1, a_2, \dots, a_n$ are numbers that are indexed by their subscripts, and the symbol above is short for the iterated sum

$$a_1 + a_2 + \dots + a_n.$$

- (1) Suppose that we have $a, b \in \mathbb{Z}$.
- (a) If a, b each leave a remainder of 1 after dividing by 4, what can you say about the remainder that ab leaves when divided by 4?
 - (b) If a, b each leave a remainder of 3 after dividing by 4, what can you say about the remainder that ab leaves when divided by 4?

- (2) Use the previous problem to show that there are infinitely many primes (such as 3, 7, 11, 19, ...) that leave a remainder of 3 after dividing by 4

Hint: This uses a variation of the construction used in Euclid's proof of the infinitude of all primes: If $p_1, \dots, p_r$ is a finite set of primes, consider the prime factorization of $N = 4p_1 \cdots p_r - 1$.

- (3) Prove the following formulas using induction:

(a)

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6};$$

(b)

$$\sum_{k=1}^n k^3 = \frac{n^2(n+1)^2}{4}.$$

- (4) Prove that $2^{2n+1} + 1$ is a multiple of 3 for every $n \in \mathbb{N}$.

- (5) Prove that for every $n \in \mathbb{N}$, we have

$$\sum_{k=1}^n \frac{1}{k^2} \leq 2 - \frac{1}{n}.$$

- (6) Use l'Hôpital's rule and induction to prove that

$$\lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0$$

for all $n \in \mathbb{N}$. You may assume that the equality is true when $n = 0$.